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Beyond Carbs:

ML-Driven Culinary Equity for T1D

PROJECT HIGHLIGHTS

Beyond Carbs:

Addressing the 80% failure rate caused by standard T1D therapy.

ML Wrapper:

Engineering FCR/PCR ratios to predict non-linear, digestive delays.

Bias Discovery:

Uncovered how AI fails the diverse majority.

I. The Narrative Challenge

Research Question:

To what extent can a ML "Wrapper" identify high-risk meals where standard insulin math (PID/MPC) fails?

Lived Experience:

11 years of lived Type 1 Diabetes experience drive this paradigm shift.

The Impact Gap:

Only 14–27% of U.S. patients meet glycemic targets, proving current carb-centric, 'calculator-style' technology is insufficient.

III. Technical Depth & Model Logic

1. Algorithm: Random Forest Classification

Ensemble Selection:

Utilizes Majority Voting to identify patterns across 386k+ self-reported logs, chosen for its resilience to non-linear physiological variance.

Optimization Goal:

Transitioning from reactive "calculators" to proactive predictors for complex, culturally diverse meals.

2. Engineering "Culinary Equity" Features

Custom FCR & PCR ratios capture the inter-macro relationships (Fat/Protein's impact on digestion & insulin timing) that standard PID/MPC math ignores.

FCR = Fat_g/Carbs_g

&

PCR = Protein_g/Carbs_g

IV. Algorithmic Bias

The 1.000 ROC-AUC:

Near-perfect ROC-AUC performance reveals a "Healthy Patient Trap," where training on idealized, low-carb clinical data creates a common signal of model overfitting.

Bias Discovery:

High accuracy masks a failure to generalize to the diverse, complex meals representative of the global majority.

Moving Forward:

Proving that for true culinary equity, AI/ML must be evaluated on its failure to support non-standard diets.

II. Technical Foundation

Dataset (T1D-UOM):

A 16-patient longitudinal multimodal study from the Manchester CS Coordinated Study.

High-Resolution Scale:

Spanning 386,564+ rows of high-frequency CGM, insulin, and nutrition logs.

Integrity & Cleaning:

Mitigated Measurement Bias via threshold checks to filter erroneous entries (e.g., impossible 500g fat logs).

V. Results & Empirical Analysis

V1: High Risk vs Low Risk

V3: Carbs vs Risk

V4: FCR vs Risk

V5: Top 10 Features

FINAL MODEL REPORT

	precision	recall	f1-score
0	1.00	1.00	1.00
1	1.00	1.00	1.00
accuracy			1.00
macro avg	1.00	1.00	1.00
weighted avg	1.00	1.00	1.00

ROC-AUC Score: 1.000

HYPOTHESIS CHECK

NOTE: Carbs (0.185) > FCR (0.015)

Evidence of Overfitting:

V1 and V5 (8-to-1 imbalance) prove the 1.000 ROC-AUC is a byproduct of skewed data, not clinical success.

Logic Paradox:

V3/V4 audits and the Hypothesis Check (Carbs 0.185 > FCR 0.015) prove the model is acting as a "glorified calculator," ignoring the multi-macro features required for Culinary Equity.

VI. Next Steps & Vision

Scaling "Culinary Equity":

Retraining on diverse dietary datasets to eliminate Carb-Centric Bias.

The ML Wrapper MVP:

Developing a simple "4-Input" (Macros + Glucose) MVP to visualize the User Journey and provide proactive risk signaling for complex meals.

Stage

Standard "Carb-Centric" Flow

ML-Driven "Culinary Equity" Flow

Input Data

Carbohydrates Isolated

Logic

Reactive PID/MPC "Calculator"

Physiology

Ignores Protein/Fat "Traffic Jams"

Outcome

Glycemic Failure: High-risk spikes

Multi-Macro FCR/PCR Ratios

Proactive Random Forest Ensemble

Decodes Complex Digestive Delays

Culinary Equity: Proactive Risk Mitigation

Open-Source Integration:

Deploying the FCR/PCR logic as a "Safety Layer" plugin for artificial pancreas systems like AndroidAPS and Loop.

VII. Referenced Sources

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[6] AI4ALL Ignite. (2025). Fall 2025 accelerator: Technical project & professional rubric standards.

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Link to GitHub Repository